

Fig. 7: Workspace section

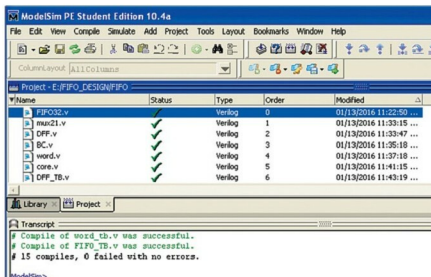


Fig. 8: Compilation window

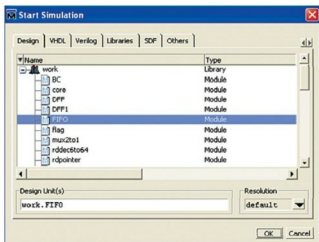


Fig. 9: Start Simulation window

Clock domain that supplies data to FIFO is often referred to as write or input logic, and clock domain that reads data from FIFO is often referred to as read or output logic. Pin diagram of FIFO is shown in Fig. 1 and its description in the table.

FIFOs are used in designs to safely pass multi-bit data words from one clock domain to another, or to control the flow of data between source and destination sides sitting in the same clock domain. If read and write clock domains are governed by the same clock signal, FIFO is said to be synchronous. If these are governed by different (asynchronous) clock signals, FIFO is said to be asynchronous. A synchronous FIFO example is described here.

Input port is controlled by write-clock and write-enable inputs. Data is written into FIFO on every rising edge when it is asserted. Output port is controlled by read-clock and read-enable inputs. Data is read from FIFO on every rising edge and read-enable is asserted.

Software program

Verilog has been used for register-transfer logic (RTL) coding and verification. The bottom-up design has been followed in this article in order to facilitate easy debugging of the various modules.

ModelSim is an easy-to-use yet versatile VHDL/SystemVerilog/Verilog/SystemC simulator from Mentor Graphics. It supports behavioural, register-transfer-level and gate-level modelling.

To start FIFO design simulation, install ModelSim V10.4a on a Windows PC and follow the steps mentioned below.

1. Start ModelSim from the desktop; you will see ModelSim 10.4a dialogue window.
2. Create a project by clicking Jump Start on Welcome screen.

or stacks so that the oldest request is handled first. In hardware, it is either an array of flops or read/write memory that stores data from one clock domain and on request supplies the same data to other clock domains following FIFO logic.